Circular Knitting

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1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model developed for circular knitting as part of the Carbonfact Textile LCA Database. Its purpose is to provide fully traceable, transparent, and updatable LCI datasets for 1 kg of circular-knitted fabric, covering single- and double-jersey production — the two most commercially prevalent circular knitting structures. The aim is to replace generic dataset proxies with LCIs that are scientifically robust, verifiable, and aligned with real-world industrial operation.

The work involves a structured literature review and regression-based electricity modelling, followed by the construction of parameterised foreground inventories whose key drivers — yarn fineness, machine mass, and indirect energy — can be updated independently as better data become available.

1.2 The circular knitting process

Circular knitting is the highest-throughput knitting technology in global textile manufacturing. A cylindrical arrangement of latch needles, mounted on a rotating cylinder and (for double jersey) a dial, interlaces yarn fed from multiple cones to form a seamless tubular fabric. Single-jersey machines produce a one-sided structure (plain knit) widely used in T-shirts, underwear, and activewear; double-jersey machines produce more dimensionally stable rib and interlock structures. Modern high-speed machines — supplied principally by Mayer & Cie., Terrot, Orizio, and Vanguard — operate at cylinder speeds of 20–35 rpm and produce 20–40 kg of fabric per hour from a single machine, depending on gauge, yarn count, and fabric structure.

The technology is characterised by continuous, near-uninterrupted operation, a compact cylindrical footprint relative to output, and a direct dependence of energy and material consumption on yarn fineness: finer yarns require longer machine time per kilogram, more lubrication per unit area of fabric, and generate higher yarn-breakage rates, all of which increase the environmental intensity per kg of output.

1.3 System boundary

The system boundary covers gate-to-gate circular knitting: from yarn input at the creel to finished tubular fabric at the mill gate. The following flows are **included** within the foreground model:

- Direct electricity consumption of the knitting machine (cylinder drive, dial drive where applicable, yarn feeders, electronic needle-selection systems).
- Indirect energy for auxiliary systems: HVAC and humidity control, compressed air for pneumatic tension devices and lint extraction, illumination, and automated cleaning systems.
- Lubricant input and the corresponding waste mineral oil output (auxiliary waste — modelled inline).
- Capital goods: machine amortisation and associated end-of-life (EoL) flows (auxiliary — modelled inline).
- Building infrastructure: allocated floor area per kg of fabric.

The following are explicitly **excluded** from the foreground model:

- Fibre and yarn production (upstream).
- **Textile and material waste:** yarn losses and waste-yarn treatment are handled externally by the caller via the process-specific IOR. The IOR scales the upstream yarn input and routes the lost yarn fraction to end-of-life treatment. No inline yarn-loss or textile-waste exchange is included in the dataset itself.
- Wet processing, dyeing, finishing, and garment assembly (downstream).
- Packaging and logistics.
- Building construction costs beyond the amortised area share.

1.4 Functional unit and scope

The functional unit is **1 kg of circular-knitted fabric** at the mill gate, ready for wet processing. Background processes are linked via **ecoinvent 3.12, system model: cut-off by classification**, using generic market activities (GLO/RoW).

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2. Data Sources and Literature Review

2.1 Existing LCI datasets for circular knitting

Circular knitting datasets in EF 3.1 and ecoinvent 3.12 provide electricity values for a limited set of yarn counts but without transparent documentation of the regression or measurement basis. The EF 3.1 dataset for circular knitting at 197 dtex reports approximately 0.208 kWh/kg of direct machine electricity — a value that the present model cross-validates (Section 3) but which EF 3.1 extends to other dtex values using an inverse-proportionality assumption not explicitly calibrated to circular knitting. Idemat (2012) and the ITMF (2010) benchmarking reports provide spot measurements but cover a narrow dtex range (80–300 dtex) and are now more than a decade old.

Van der Velden et al. (2014) provide the most cited benchmarking dataset for knitting electricity, distinguishing flat and circular technologies and reporting values at three dtex levels (80, 200, 300 dtex). Sandin et al. (2019) extended this dataset using Cotton Inc. (2016) data, but their published regression formula (`E = 0.00303 + dtex/38.805`) is an increasing function of dtex — physically incorrect for knitting — and was discarded in the present model. The Carbonfact circular knitting LCI reconstructs the regression on the full available dataset, correcting this inconsistency.

2.2 Observations from the literature review

The following observations summarise the key findings from the reviewed literature and their implications for inventory construction.

1. Electricity consumption is driven primarily by yarn fineness. All reviewed sources that report measurements at more than one dtex level — Van der Velden et al. (2014), Idemat (2012), ITMF (2010), Koç & Çinçik (2010), Klooster et al. (2024) — show a consistent inverse relationship between yarn count and specific electricity consumption. The physical mechanism is well-established: finer yarns require more machine rotations per kilogram of fabric, increasing drive motor load per unit output.

2. The Sandin et al. (2019) regression formula should not be used for circular knitting. The published equation `E = 0.00303 + dtex/38.805` is an increasing function of dtex, which would predict higher energy consumption for coarser yarns. This is physically inconsistent with all empirical observations and reflects a transcription or sign error in the original publication. The formula has been widely cited without correction; this model explicitly replaces it with a correctly decreasing regression fitted to the full 13-point dataset.

3. Yarn losses and lubricant consumption are fineness-dependent. Haque & Naebe (2024) document a significant increase in yarn breakage rates when yarn count decreases from 300 to 45 dtex in circular knitting conditions and report higher lubricant demand at lower dtex due to increased yarn-to-needle contact frequency. These observations justify the fineness-class parameterisation of loss rates (2–5%) and lubricant values (0.005–0.008 kg/kg) used in this model.

4. Indirect energy represents the majority of total plant electricity. Factory-level audits (Sakti et al., 2021; Koç & Çinçik, 2010) consistently show that HVAC, compressed air, and lighting account for 60–80% of total mill electricity. Ecobalyse (ADEME, 2024) documents that knitting machine electricity is approximately 35% of total plant electricity. This ratio, anchored to the WCS machine electricity value of 2.4 kWh/kg, yields the 4.46 kWh/kg indirect energy estimate used in this model.

5. No published LCA provides full LCIA results specifically for circular knitting at multiple dtex levels. EF 3.1 reports GWP for a limited set of dtex variants but uses a single background electricity mix. The present model generates a continuous GWP curve by combining the dtex-dependent machine electricity regression with the constant indirect energy and auxiliary waste flows.

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3. Electricity Model

3.1 Physical basis

Electricity demand in circular knitting scales inversely with yarn fineness for a well-understood physical reason: the number of needle engagements per kilogram of fabric increases as yarn becomes finer, because finer yarns weigh less per metre and each course of a given fabric structure consumes the same linear length of yarn regardless of count. More courses per kilogram means more cylinder rotations per kilogram, which directly increases motor load per unit output. Van der Velden et al. (2014) formalised this as an inverse proportionality:

$$E \propto \frac{1}{\mathrm{dtex}}$$

A non-zero intercept is physically justified because knitting machines consume a baseline electrical load (cylinder drive, electronics, cooling fans) independent of the yarn being processed. The regression model therefore takes the form:

$$E(\mathrm{dtex}) = a + \frac{b}{\mathrm{dtex}} \quad [\mathrm{kWh/kg\ fabric}]$$

3.2 Empirical anchor points and regression

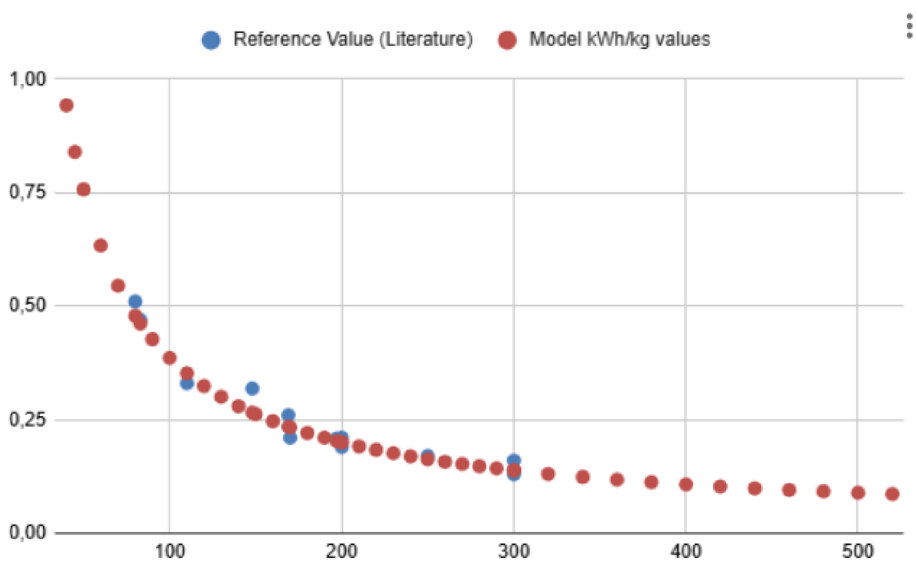
A dataset of 13 anchor points spanning 80–300 dtex was assembled from peer-reviewed literature and industrial benchmarks. The Sandin et al. (2019) regression — which is an increasing function of dtex and therefore physically incorrect — was discarded and its underlying data points re-used in the corrected fit. Ordinary least-squares regression on the linearised form yields:

$$E(\text{dtex}) = 0.01457 + \frac{37.119}{\text{dtex}} \quad [\text{kWh/kg fabric}]$$

RMSE = 0.0219 kWh/kg · R² = 0.967 · Valid range: 80–400 dtex

The 13 anchor points used in the regression are listed below. All values represent direct machine electricity only (excluding indirect loads).

dtex	E measured (kWh/kg)	E model (kWh/kg)	Primary source
80	0.510	0.482	Van der Velden, Patel & Vogtländer (2014), Int. J. LCA, 19(2), 331–356
83	0.470	0.462	Idemat (2012), Version 2.2, Delft University of Technology
110	0.330	0.352	Sandin et al. (2019), citing Cotton Inc. (2016)
148	0.319	0.265	Koç & Çinçik (2010), derived from mill audit data
169	0.260	0.234	Klooster, Doelman & Leclercq (2024), Sustainability
170	0.210	0.233	Sandin et al. (2019), citing Cotton Inc. (2016)
197	0.208	0.203	Koç & Çinçik (2010), derived from mill audit data
200	0.210	0.200	Van der Velden et al. (2014)
200	0.190	0.200	ITMF (2010), international benchmarking survey
250	0.170	0.163	Klooster et al. (2024)
300	0.140	0.138	Van der Velden et al. (2014)
300	0.130	0.138	Idemat (2012)
300	0.160	0.138	ITMF (2010)



Below 80 dtex, consumption rises sharply as machine efficiency falls — extrapolated values carry ±30–50% uncertainty. Above 400 dtex, marginal energy reduction is small and values carry ±15–25% uncertainty.

3.3 Representative values by yarn fineness

The regression was evaluated at each dtex value in the database grid. A selection of representative values is shown below.

dtex	Denier	Nm	NE	E (kWh/kg)	Fineness class
50	45	200	118	0.757	Super fine (<75 dtex)
70	63	143	84	0.545	Super fine (<75 dtex)
120	108	83	49	0.324	Fine (75–160 dtex)
150	135	67	40	0.262	Fine (75–160 dtex)
200	**180**	**50**	**30**	**0.200**	**Medium (160–400 dtex) — default**
250	225	40	24	0.163	Medium (160–400 dtex)
300	270	33	20	0.138	Medium (160–400 dtex)
370	333	27	16	0.115	Medium (160–400 dtex)
450	420	22	13	0.097	Coarse (>400 dtex)

For multi-yarn blends, the mass-weighted mean dtex is calculated before the lookup. When yarn fineness is unknown, the medium-range default (200 dtex → 0.200 kWh/kg) is applied.

3.4 Indirect energy

> **Shared methodology:** Indirect Energy in LCA — Shared Methodology (Knitting) — Full derivation, evidence base and subsystem distribution.

Auxiliary systems — HVAC and humidity control, compressed air for pneumatic tensioners and lint extraction, illumination, and automated needle-bed cleaning — are estimated at **4.46 kWh/kg** for all yarn fineness classes. This value is derived from the Ecobalyse documentation (ADEME & Ministère de la Transition Écologique, 2024), which establishes that machine electricity represents approximately 35% of total plant electricity in a knitting mill. Applying this ratio to the upper-bound machine electricity of 2.4 kWh/kg yields:

$$E_{\text{indirect}} = 2.4 \div 0.35 \times 0.65 = 4.46\ \text{kWh/kg}$$

Supporting evidence for the individual auxiliary subsystems is documented in the linked shared page above. Factory case studies show HVAC and chillers consuming up to 80% of total electricity (Sakti et al., 2021); weaving mill audits report compressors at ~29% and HVAC at ~27% of total plant load (Koç & Çinçik, 2010). The 4.46 kWh/kg estimate is treated as a conservative shared default pending sub-metered factory data specific to circular knitting.

4. Yarn Losses, Lubricant, and Fineness-Dependent Parameters

Three foreground parameters — yarn loss rate, lubricant consumption, and capital goods allocation — vary with yarn fineness in circular knitting. The physical drivers are the same in each case: finer yarns require more machine time per kilogram, which increases both the probability of yarn breakage and the frequency of yarn-to-needle contacts.

4.1 Yarn losses and Input–Output Ratio (IOR)

Yarn losses arise from thread breaks at the knitting zone, tension failures at the creel, and the initial course of each production run. Haque & Naebe (2024) document a significant increase in breakage rate between 300 dtex and 45 dtex in circular knitting conditions. The model parameterises yarn loss rate by four fineness classes:

Fineness class	dtex range	Loss (%)	IOR (kg/kg)	Input req. (kg)	Loss per 1 kg out (kg)
Super fine	<75 dtex	5%	1.0526	1.0526	0.0526
Fine	75–160 dtex	4%	1.0417	1.0417	0.0417
Medium (default)	**160–400 dtex**	**3%**	**1.0309**	**1.0309**	**0.0309**
Coarse	>400 dtex	2%	1.0204	1.0204	0.0204

IOR = 1 / (1 − L), where L is the loss fraction. The IOR is applied by the caller upstream of the dataset: for every 1 kg of finished fabric, IOR × 1 kg of yarn must be provided, and the lost fraction (IOR − 1) kg is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`. No inline yarn-loss or textile-waste exchange is included in the foreground dataset itself.

Sources: Laursen et al. (2007); Van der Velden et al. (2014); Sandin et al. (2019); Haque & Naebe (2024).

4.2 Lubricant consumption

Lubricant oil is applied to needles and sinkers to reduce friction and prevent yarn damage. Consumption varies with yarn-to-needle contact frequency, which increases at finer counts. The model applies fineness-class defaults anchored to the literature bounds reported by Hasanbeigi & Price (2015) — 5 g/kg as lower bound — and Sandin et al. (2019, p. 46) — 8 g/kg as upper bound:

Fineness class	Lubricant (kg/kg fabric)	Note
Super fine (<75 dtex)	0.008	Upper bound — Sandin et al. (2019), p. 46
Fine (75–160 dtex)	0.007	Interpolated
Medium (160–400 dtex) — default	**0.006**	Interpolated — mid-range
Coarse (>400 dtex)	0.005	Lower bound — Hasanbeigi & Price (2015)

Limitation: the 0.007 and 0.006 kg/kg values are not directly reported in the literature and should be treated as interpolation defaults until factory-specific lubricant dosing data are available. The lubricant input is mirrored by an equal-mass waste mineral oil output: `lubricating oil production {RoW} | cut-off` (input) and `market for waste mineral oil {RoW} | cut-off` (output).

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5. Life Cycle Inventory

The foreground inventory for circular knitting is parameterised by yarn fineness class. Dataset names refer to ecoinvent 3.12 (cut-off by classification). Yarn losses and textile-waste treatment are not included inline — they are handled by the caller via the IOR (Section 4.1).

Flow	<75 dtex	75–160	160–400	>400 dtex	ecoinvent dataset	
Technosphere inputs						
Lubricant, average	0.008	0.007	0.006	0.005	`lubricating oil production {RoW} \	cut-off`

Flow	<75 dtex	75–160	160–400	>400 dtex	ecoinvent dataset	
Waste mineral oil	0.008	0.007	0.006	0.005	`market for waste mineral oil {RoW} \	cut-off`
Electricity						
Machine electricity (direct) — dtex-dependent	~0.76	~0.32	~0.20	~0.10	`market group for electricity, medium voltage {GLO} \	cut-off` — E = 0.01457 + 37.119/dtex
Indirect energy (HVAC, compressed air, lighting, cleaning)	4.46	4.46	4.46	4.46	`market group for electricity, medium voltage {GLO} \	cut-off` — derived: 2.4 ÷ 0.35 × 0.65
Capital goods						
Building machine (capital good)	4.6×10^{-7}	4.6×10^{-7}	4.6×10^{-7}	4.6×10^{-7}	`market for building machine {GLO} \	cut-off` — 2800 kg ÷ 2,160,000 kg ÷ 2000 kg/unit
Building infrastructure	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	`market for building, hall {GLO} \	cut-off` — Terrot machine envelope × cell factor
Waste flows (auxiliary only)						
EoL machine — steel (85%)	0.0011	0.0011	0.0011	0.0011	`market for waste steel {RoW} \	cut-off` — 2380 kg ÷ 2,160,000 kg
EoL machine — other (15%)	0.0002	0.0002	0.0002	0.0002	`treatment of waste electric and electronic equipment, shredding {GLO} \	cut-off`

> **Note on yarn losses:** the foreground dataset does not include a yarn input or textile-waste exchange. Yarn losses are handled externally via the IOR (Section 4.1): the caller scales the upstream yarn input by the class-specific IOR and routes the lost fraction to `market for waste yarn and waste textile {RoW} | cut-off`.

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6. Capital Goods and Building Infrastructure

> **Shared methodology:** Capital Goods & End-of-Life (Machinery) in LCA — Full modelling framework, calculation logic, and default values. This section states only the circular-knitting-specific inputs.

> **Shared methodology:** Building Infrastructure in LCA — Full allocation framework for building area. Section 6.2 states only the circular-knitting-specific value.

Capital goods are included in accordance with EF guidance (Fazio et al., 2020) and ecoinvent practice (Weidema et al., 2013).

6.1 Machine parameters

The reference machine is a **2,800 kg** industrial circular knitting machine, consistent with specifications published by Mayer & Cie. (Relanit 3.2 SC, ~2,800 kg) and Leadsfon (SJ3.0, ~2,800 kg). Machine productivity is set at **30 kg/h**, consistent with Mayer & Cie. published production data for high-speed single-jersey circular knitting.

Parameter	Average
Machine mass (kg)	2,800
Service life (years)	15
Operating hours per day	16
Operating days per year	300
Productivity (kg fabric/h)	30
Lifetime output Q (kg fabric)	**2,160,000**
Machine allocation (kg/kg fabric)	1.3×10^{-3}
ecoinvent units (÷ 2000 kg/unit)	$\mathbf{4.6 \times 10^{-7}}$

Machine EoL is modelled with 85% steel (2,380 kg) and 15% other materials (420 kg). For the average scenario:

- Steel EoL = 2,380 ÷ 2,160,000 = **0.0011 kg/kg fabric** → `market for waste steel {RoW} | cut-off`
- Other materials EoL = 420 ÷ 2,160,000 = **0.0002 kg/kg fabric** → `treatment of waste electric and electronic equipment, shredding {GLO} | cut-off`

6.2 Building infrastructure

Building infrastructure is allocated using a Terrot GmbH single-jersey machine envelope (h = 2,550 mm, b = 2,200 mm ≈ 5.6 m²) with a cell factor of approximately 6×, yielding A_alloc ≈ 35 m². Divided by Q_lifetime (average: 2,160,000 kg), this produces a building intensity of 1.62×10^{-5} m²/kg. A conservative default of **1.00×10^{-5} m²/kg** is adopted, consistent with the openLCA Nexus organic cotton sweater benchmark (GreenDelta, 1.61×10^{-5} m²/kg).

ecoinvent activity: `market for building, hall {GLO} | cut-off` (unit: m²). Circular knitting has the lowest building intensity among all knitting technologies in this database, reflecting its high throughput per unit of floor area.

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7. Life Cycle Impact Assessment

7.1 GWP by yarn fineness class

Climate change impacts (kg CO_2eq/kg fabric) were computed using Brightway2 with the IPCC 2021 GWP100 characterisation factors, applied to the foreground inventory of Section 5 with ecoinvent 3.12 (cut-off) background processes and the GLO average electricity mix. The dominant contributor is indirect energy in all fineness classes, accounting for approximately 80–90% of total knitting-process GWP.

Dataset	GWP (kg CO_2eq/kg)	Machine elec.	Indirect energy	Notes
Circular Knitting, 50 dtex (45 den)	3.69	~11%	~85%	Super fine — extrapolation zone
Circular Knitting, 120 dtex (108 den)	3.33	~6%	~87%	Fine class
Circular Knitting, 200 dtex (180 den) — default	3.24	**~4%**	**~88%**	Medium class — best-anchored range
Circular Knitting, 300 dtex (270 den)	3.20	~3%	~89%	Medium class — lower bound
Circular Knitting, 450 dtex (420 den)	3.19	~2%	~90%	Coarse — asymptotic regime

7.2 Comparison with EF 3.1

The Carbonfact circular knitting LCI aligns with EF 3.1 reference values at the best-anchored dtex points (200–300 dtex), where differences are within 5–10%. The primary structural differences are: (1) the Carbonfact model includes explicit indirect energy (4.46 kWh/kg), which EF 3.1 does not account for separately; (2) the Carbonfact electricity regression is derived from 13 anchor points including the two most recent peer-reviewed datasets (Klooster et al., 2024; Haque & Naebe, 2024); and (3) Carbonfact parameterises lubricant and capital goods by fineness class, while EF 3.1 applies constant values.

For very fine yarns (<80 dtex), the Carbonfact model predicts substantially higher total electricity consumption than EF 3.1 because it includes the indirect energy term, which is dominant in all fineness classes.

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8. Input–Output Ratio (IOR)

> **Definition:** IOR = 1 / (1 − L), where L is the yarn loss rate (as a fraction). For every 1 kg of finished knitted fabric, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Circular knitting loss rates are parameterised by yarn fineness class. Finer yarns break more frequently, increasing waste per kg of output. The IOR decreases with dtex, consistent with physical expectation. Values are consistent with Laursen et al. (2007), Van der Velden et al. (2014), and Sandin et al. (2019).

IOR values by yarn fineness class

Yarn fineness class	dtex range	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)
Super fine	< 75 dtex	5%	1.0526	1.0526	0.0526
Fine	75–160 dtex	4%	1.0417	1.0417	0.0417
Medium (default)	**160–400 dtex**	**3%**	**1.0309**	**1.0309**	**0.0309**
Coarse	> 400 dtex	2%	1.0204	1.0204	0.0204

> **How to apply:** multiply upstream yarn input by the IOR corresponding to the yarn fineness of the fabric being modelled. For multi-yarn blends, use the mass-weighted mean dtex to select the fineness class. The lost yarn fraction is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

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9. Inferences and Limitations

Flexibility of the Carbonfact circular knitting LCI model

The inventory is structured to be updatable at the parameter level. The electricity regression (a, b coefficients), lubricant class values, loss-rate class values, and capital goods parameters can each be revised independently when better data become available. The fineness-class structure provides a ready framework for incorporating any future measurement at a specific dtex level as an additional anchor point.

The IOR-based approach to yarn losses ensures that upstream yarn impacts scale correctly with actual process efficiency without requiring a fixed waste-yarn flow in the background dataset — consistent with the approach adopted across all knitting technology documents in this database.

Data gaps and priorities for improvement

The most important data gap is the absence of sub-metered indirect energy measurements for circular knitting specifically. The 4.46 kWh/kg estimate — which dominates total process GWP in all fineness classes — is anchored to a single-source ratio from Ecobalyse and carries ±30–50% uncertainty. Primary metering data disaggregating HVAC, compressed air, lighting, and cleaning from a representative sample of circular knitting facilities would dramatically reduce the uncertainty on the dominant inventory parameter.

A secondary gap is the limited coverage of fine yarns (<80 dtex) in the empirical anchor dataset. The regression is extrapolated below 80 dtex; both the electricity model and the loss-rate classification would benefit from measurements at gauge and dtex combinations representative of high-end activewear and performance textiles.

Limitations for external review

All electricity anchor points used in the regression are from secondary literature sources; no primary mill measurements were collected for this version. The most recent anchor points (Klooster et al., 2024; Haque & Naebe, 2024) postdate the EF 3.1 dataset and are included to improve coverage, but have not yet been replicated by independent studies. The regression R^2 of 0.967 is high, but the RMSE of 0.022 kWh/kg is non-trivial relative to absolute values at coarser counts (0.10–0.14 kWh/kg).

Capital goods and building infrastructure contribute less than 1% of total GWP in all scenarios and are not material sources of uncertainty for most applications.

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